FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2024
FULL TEST – VIII
PAPER –1
TEST DATE: 05-05-2024

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. A, C

Sol. Force balance on system when raft is half submerged

$$(3M+3m+m_0)g=\frac{6B}{2} \quad \ldots(i)$$

Force balance on system when raft is fully submerged

$$(m+m_0)g=B \quad \ldots(ii)$$

and B is the buoyancy force on a completely submerged log.

Torque balance relative to the centre of the raft

$$3Mg-(3m+m_0)g=0 \quad \ldots(iii)$$

From equations (i), (ii) and (iii) we find that $\frac{M}{m}=2$ and $\frac{(3M+3m)}{m_0}=3$.

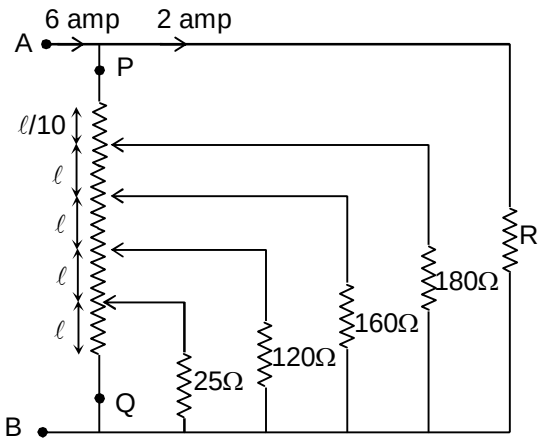
2. A, B, C, D

Sol. Equivalent resistance between P and Q $= 100\ \Omega$

So current towards $P=\frac{400}{100}=4$ ampere

Current through $R=6-4=2$ ampere

So, $R=\frac{400}{2}=200\ \Omega$



3. A, B, C, D

Sol. At any instant the total number of moles $n_1+n_2=n$ remains constant

$$\frac{n_1}{n_2}=\frac{P_1V_1}{P_2V_2}=\left(\frac{\frac{4S}{R_1}\times\frac{4}{3}\pi R_1^3}{\frac{4S}{R_2}\times\frac{4}{3}\pi R_2^3}\right)=\left(\frac{R_1}{R_2}\right)^2=\frac{1}{4}$$

$$\frac{P_1V_1}{RT}+\frac{P_2V_2}{RT}=2+8$$

$$\frac{4S}{R_0}\frac{4}{3}\pi R_0^3+\frac{4S}{2R_0}\frac{4}{3}\pi 8R_0^3=10\ RT$$

$$2(4\pi R_0^2+16\pi R_0^2)=10RT\times\frac{3}{2S}$$

$A_1+A_2=\frac{15RT}{S}$ and it is constant as T is constant

When $n_1 = 1$; $n_2 = 9$

$\frac{P_1V_1}{P_2V_2}=\frac{1}{9}=\frac{r_1^2}{r_2^2}$ or $r_2 = 3R_1$ and $40\pi\cdot R_0^2=8\pi(r_1^2+r_2^2)=8\pi(10r_1^2)$

$$r_1=\frac{R_0}{\sqrt{2}}$$

4. B

Sol. The rotation of the rocket velocity vector is associated with the displacement of the target: during the time Δt after

launch $\Delta\phi\approx\arctan\left(\frac{u\Delta t}{H}\right)\approx\frac{u\Delta t}{H}$

$$\omega=\frac{\Delta\phi}{\Delta t}=\frac{u}{H}$$

$$a=\omega v=\frac{uv}{H}$$

a is in horizontal direction and mg is in vertical direction.

$$\left|\vec{F}_m\right|=m\sqrt{g^2+\left(\frac{vu}{H}\right)^2}\ .$$

5. B

Sol. Let M is the mass of the rope and let horizontal part of the tube in the initial state is the nth part of the rope.

At equilibrium:

$m(1-n)g=kx_n$

x_n-initial elongation.

Let further elongation is x

$$(m+M)a=Mg+mg\left(1-n+\frac{x}{L}\right)-k(x_n+x)$$

Or

$$(m+M)a=Mg+\frac{mgx}{L}-kx$$

To have constant acceleration term with x has to be eliminated.

$(m+M)a=Mg$ and $mg=kL$

6. B

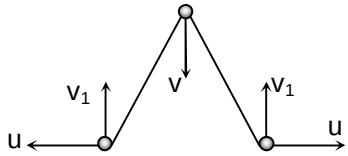
Sol. conservation of momentum:

$mv - 2mv_1 = 0$

Along the string equating velocity

$v\cos45° = -v_1\cos45° + u\sin45°$

from where $v_1 = \frac{v}{2}$ and $u = \frac{3v}{2}$.



From the law of conservation of energy:

$$\frac{mv^2}{2} + 2\frac{mv_1^2}{2} + 2\frac{mu^2}{2} = \frac{kq^2}{\ell} - \frac{kq^2}{\ell\sqrt{2}}$$

Where $k = \frac{1}{4\pi\varepsilon_0}$.

7. A

Sol. HCF of 2000 Hz and 1200 Hz = 400 Hz

2000 Hz Corresponds to second over tone or $2\frac{1}{2}$ loops

$\therefore\ \ell = \frac{5}{2}\times 8\text{ cm} = 20\text{cm}$

8. A

Sol. if object density > liquid density and if object sinks into liquid from initial floating condition then level of water fall.

(P) as initially ball was floating and later also they will float, therefore d = const. (T), h = decreases (P)

(Q) d↓ and h↓ (P, S)

(R) (Q, T)

(S) Water and balloon both will expand, therefore d↑

$$T = B = \rho_L Vg = \frac{P_0}{(1+\gamma_L\Delta T)}(1+\gamma_{air}\Delta T)g$$

as $\gamma_{air} > \gamma_L \Rightarrow$ T will increase

9. D

Sol. Use basic concept of kinematics of rolling

10. A

Sol. Potential inside the solid ball (r < R)

$$V = -\frac{GM}{2R^3}(3R^2 - r^2)$$

r > R

$$V = \frac{GM}{r}$$

Using superposition theorem will get desired result.

11. B

Sol. $R' = R\left[1+10^{-5}\times 10\right] \Rightarrow \frac{\Delta R}{R} = 10^{-4}$

$\frac{\Delta R}{R}\times 100 = 0.01$

$\Rightarrow$ change in radius 0.01%

(A) $K = \frac{1}{2}I\omega^2$

(B) $\omega = \frac{V}{R}$

(C) $L = I\omega$

(D) $I \propto R^2$

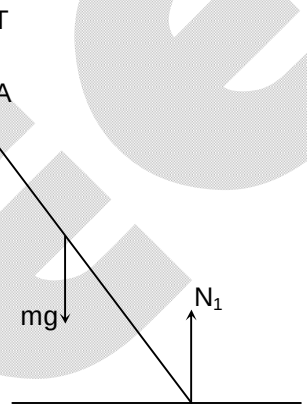
SECTION – B

12. 1

Sol. Initially torque balane about point A.

$N_1 L\sin\theta = mg\frac{L}{2}\sin\theta$

$N_1 = \frac{mg}{2}$



Just after cutting the string

Torque equation about Instantenous axis of rotation B

$\alpha = \frac{(mg)\frac{L}{2}\sin\theta}{\frac{mL^2}{12} + m\left(\frac{L}{2}\sin\theta\right)^2}$

$\sin\theta = \frac{1}{\sqrt{3}}$

$\alpha = \frac{\sqrt{3}g}{L}$

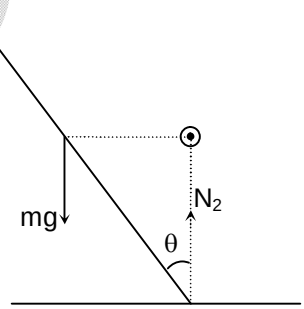
$a_y = \alpha\frac{\ell}{2}\sin\theta$

$a_y = \frac{g}{2}$

$mg - N_2 = ma_y$

$N_2 = mg/2$

$\frac{N_1}{N_2} = 1$



13. 4

Sol. force balance on 1 along y axis:

$-2T\cos\alpha + 2F_{12}\cos\alpha + F_{11} = 0$ …(i)

where α is the angle between the right side and the vertical,

T is the tension force of the thread,

F_{11} is the force of repulsion from the charge at the other end of the diagonal,

F_{12} is the force of repulsion from the charge at the other end of the side of the rhombus.

force balance on 2 along x-axis:

$-2T\sin\alpha + 2F_{12}\sin\alpha + F_{22} = 0$ …(ii)

where F_{22} is the force of repulsion from the charge at the other end of the diagonal.

Dividing (1)by cosine and (2) by sine,

we get:

$\frac{F_{11}}{\cos\alpha} = \frac{F_{22}}{\sin\alpha} \Leftrightarrow \frac{kq^2}{d_1^2\cos\alpha} = \frac{k(Nq)^2}{d_2^2\sin\alpha}$

From geometry, the ratio of diagonals $\frac{d_2}{d_1} = \tan\alpha$. Substituting in (2) we get:

$$\tan\alpha = N^{2/3} \Rightarrow \frac{d_2}{d_1} = N^{2/3} = 4$$

14. 1

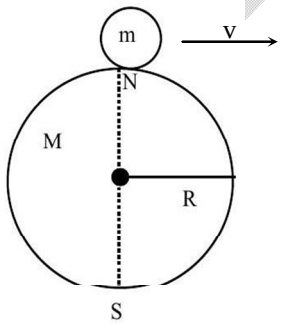
Sol. $\frac{R \cdot 2R}{R + 2R} = \frac{2}{3}R$

Thus, the total current:

$$I = \frac{E}{\frac{2}{3}R} = \frac{2}{\frac{2}{3}\cdot 3} = 1\,A$$

15. 2

Sol. When the centre of the satellite is at a distance y from the centre of the earth, By conservation of energy



$$\frac{1}{2}mv^2 - \frac{GMm}{R+r} = \frac{1}{2}mv_1^2 - \frac{GMm}{y}$$

Since, r << R

$$\frac{1}{2}mv^2 - \frac{GMm}{R} = \frac{1}{2}mv_1^2 - \frac{GMm}{y}$$

Also by conservation of angular momentum at the for that distance

$$mvR = myv_1 = \frac{vR}{y}$$

$$\therefore \frac{1}{2}mv^2 - \frac{GMm}{R} = \frac{1}{2}mv_1^2 - \frac{GMm}{y} = \frac{1}{2}\frac{mv^2R^2}{y^2} - \frac{GMm}{y}$$

$$\Rightarrow y = \frac{v^2R^2}{2GM - v^2R}$$

If $Rv^2 = 2GM \Rightarrow y = \infty$

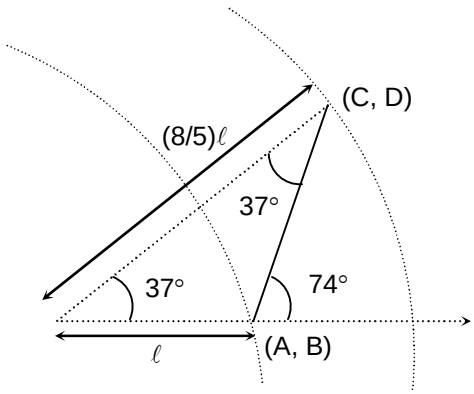
$\therefore$ Satellite will escape to infinity

16. 235

Sol. Flux passing through the loop ABCD

$$\phi = \int_{\ell}^{(8/5)\ell} \frac{\mu_0 I}{2\pi r}\ell dr = \frac{\mu_0 I\ell}{2\pi}\ell n\left(\frac{8}{5}\right)$$

$$\phi = \left(\frac{\mu_0\ell}{\pi}\right)\left(\frac{0.47\times 10}{2}\right) = 2.35\left(\frac{\mu_0\ell}{\pi}\right)$$



17. 3

Sol. $\vec{F} = q(v_x\hat{i} + v_y\hat{j}) \times c\left(1+\frac{x}{L}\right)(\hat{k})$

$\vec{F} = qcv_x\left(1+\frac{x}{L}\right)(-\hat{j}) + qcv_y\left(1+\frac{x}{L}\right)(\hat{i})$

$F_y = -qcv_x\left(1+\frac{x}{L}\right)$

$m\frac{dv_y}{dt} = -qc\frac{dx}{dt}\left(1+\frac{x}{L}\right)$

$m\int_{u_{y1}}^{u_{y2}} dv_y - qc\int_0^L\left(1+\frac{x}{L}\right)dx$

$m\left(u_{y_2} - u_{y1}\right) = -qcL - \frac{qcL}{2} = -\frac{3}{2}qcL$

$m(v\cos\theta_2 - v\cos\theta_1) = -\frac{3}{2}qcL$

$2mv(\cos\theta_2 - \cos\theta_1) = 3qcL$

Chemistry

PART – II

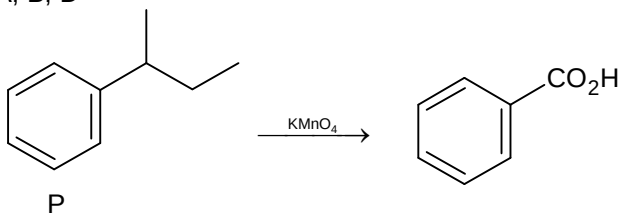
SECTION – A

18. A, B, D

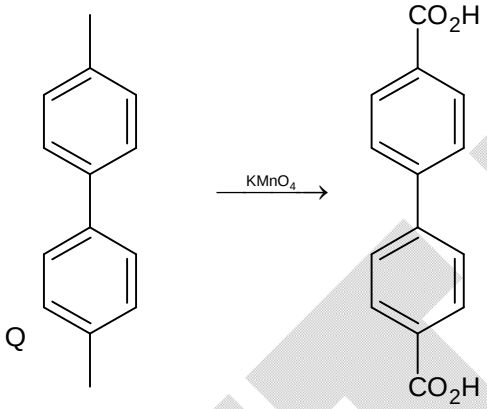
Sol. The most probable oxidation states of Cr and Mo are +2, +3 and +6.
Mo(VI) and W(VI) are more stable than Cr(VI).
CrO_3 is an acidic oxide.

19. A, B, D

Sol.



P

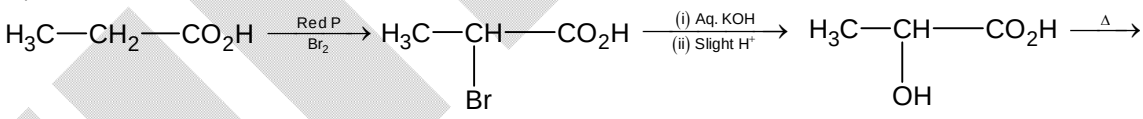


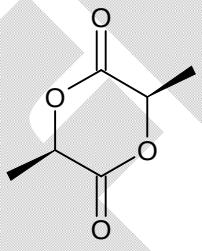
Q

Both P and Q are weakly activating towards EAS reactions.

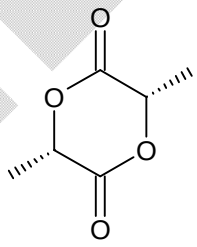
20. B, C

Sol.

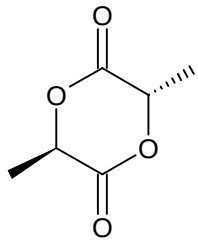




D,D-lactide
optically active



L,L-lactide
optically active



Meso-lactide
optically inactive

21. C

Sol. $\alpha = \sqrt{\frac{K_a}{C}} = \sqrt{\frac{8\times10^{-5}}{0.2}}$

$= \sqrt{4\times10^{-4}}$

$= 2\times10^{-2}$

= 0.02

$i = 1+\alpha$

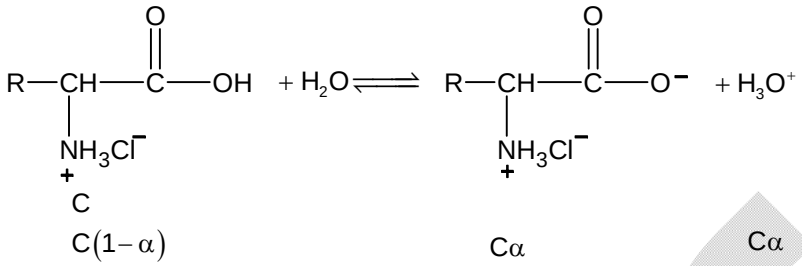
$\pi = i\times CRT$

$= (1.02)\times 0.2\times 0.0821\times 300$ atm

= 5.024 atm

22. B

Sol.



C

$C(1-\alpha)$ $C\alpha$ $C\alpha$

$pH = 2.7; \quad \therefore [H^+] = 2\times10^{-3}$

$C = \dfrac{0.025\times1000}{250} = 0.01\,M$

$[H^+] = \sqrt{K_{a_1}C}$

$2\times10^{-3} = \sqrt{K_{a_1}\times0.01}$

$4\times10^{-6} = K_{a_1}\times0.01$

$K_{a_1} = 4\times10^{-4}$

$pK_{a_1} = 4-\log4$

= 4 – 2log2

= 3.4

$K_{a_2} = 8\times10^{-8}$, $pK_{a_2} = 8-\log8$

= 8 – 3 log2

= 7.1

$pHi = \dfrac{1}{2}(pK_{a_1}+pK_{a_2})$

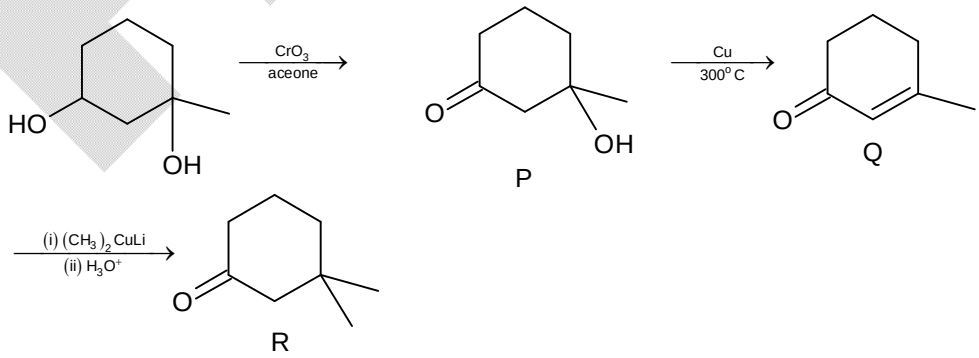
$= \dfrac{1}{2}(3.4+7.1)$

$= \dfrac{1}{2}\times10.5$

= 5.25

23. B

Sol.



24. D

Sol. Borax bead or microcosmic bead test is performed by coloured cations only and Zn^{2+} is colourless due to the absence of unpaired electron, so it cannot be detected by the borax bead or microcosmic bead test.

25. B

Sol. $2KClO_3 \xrightarrow{\Delta} 2KCl + 3O_2$

$2KMnO_4 \xrightarrow{\Delta} K_2MnO_4 + MnO_2 + O_2$

$2PbO_2 \xrightarrow{\Delta} 2PbO + O_2$

$4K_2Cr_2O_7 \xrightarrow{\Delta} 4K_2CrO_4 + 2Cr_2O_3 + 3O_2$

All of them are strong oxidizing agents,

$KClO_3$ and PbO_2 used in safety matches.

26. A

Sol. Amides and aryl amines are weaker base than alkyl amines.

$H_3C-\underset{\underset{O}{\|}}{C}-NH_2 \xrightarrow[NaOH]{\Delta} CH_3COONa + NH_3 \uparrow$

Hydrazine was used as a component of rocket fuel.

27. B

Sol. $P_4 + 3NaOH + 3H_2O \longrightarrow PH_3 + 3NaH_2PO_2$

$2P + 2NaClO_2 + 2H_2O \longrightarrow Na_2H_2P_2O_6 + 2HCl$

$H_3PO_3 + PCl_3 + H_2O \longrightarrow H_4P_2O_5 + HCl$

$5H_3PO_4 + POCl_3 \longrightarrow 3H_4P_2O_7 + 3HCl$

28. C

Sol. Absorption and desorption are endothermic process.

Physisorption is multimolecular.

SECTION – B

29. 4

Sol. $4r = \sqrt{3}a_{bcc}$; $4r = \sqrt{2}a_{fcc}$

$\therefore \sqrt{3}a_{bcc} = \sqrt{2}a_{fcc}$

$$d_{bcc} = \frac{2 \times A}{N_A \times (a_{bcc})^3} \quad \text{.... (1)}$$

$$d_{fcc} = \frac{4 \times A}{N_A \times (a_{fcc})^3} \quad \text{... (2)}$$

$(1) \div (2)$

$$\frac{d_{bcc}}{d_{fcc}} = \frac{2(a_{fcc})^3}{(a_{bcc})^3 \times 4}$$

$$d_{fcc} = d_{bcc} \times \left(\frac{a_{bcc}}{a_{fcc}}\right)^3 \times 2$$

$$= \frac{3\sqrt{3}}{\sqrt{2}} \times \left(\frac{\sqrt{2}}{\sqrt{3}}\right)^3 \times 2$$

= 4

30. 8

Sol. 160 kg SO_3 produces 128 kg SO_2 and 32 kg O_2

$\therefore$ 32 kg SO_3 produces 25.6 kg SO_2 and 6.4 kg O_2

So, at least 32 kg SO_3 is required to produce 6.4 kg of each product if 100% yield.

$\therefore$ For 50% yield

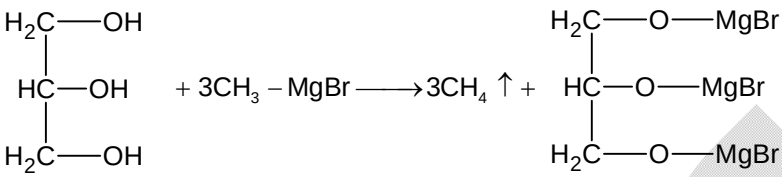
Minimum mass of SO_3 required is = 64 kg

So, x = 64

$$\therefore \frac{x}{8} = 8 \text{ kg}$$

31. 4

Sol.



Molar mass of glycerol = 92

$$\text{Moles of glycerol} = \frac{4.6}{92} = \frac{1}{20}$$

$$\text{Moles of } CH_4 \text{ produced} = \frac{3}{20}$$

$$\text{Weight of } CH_4 \text{ produced} = \frac{3}{20} \times 16$$

$$= \frac{12}{5} \text{g}$$

$$\therefore x = \frac{12}{5}$$

$$\therefore \frac{5x}{3} = \frac{12}{5} \times \frac{5}{3}$$

= 4

32. 5

Sol. In the complex $[M_{ABCDEF}]$

Total geometrical isomers = 15

All are optically active

$\therefore$ Total number of stereoisomers

x = 30.

In the complex $[M_{A_2BCDE}]$

Total geometrical isomers = 9

cis = 6; trans = 3

All cis isomers are optically active

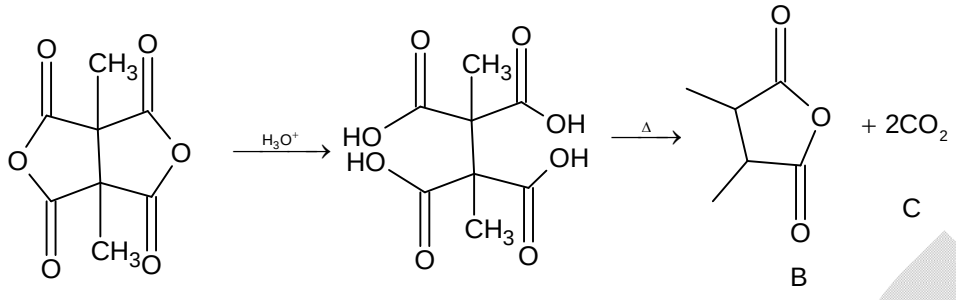
$\therefore$ Total number of stereoisomers

y = 15.

$$\therefore \frac{x-y}{3} = \frac{30-15}{3} = 5$$

33. 22

Sol.



A B C

1 mol A produce 2 mols of CO_2 (C)

$\therefore$ 0.25 mol A produce 0.5 mols of CO_2(C)

$\therefore$ Amount in g =0.5 × 44

= 22 g

34. 6

Sol. $x = 18, y = 5$

$\therefore \frac{x \times y}{15} = 6$

Mathematics

PART – III

SECTION – A

35. A, B, C, D

Sol. $C : x^2 + y^2 - 6x - 10y + 30 = 0$

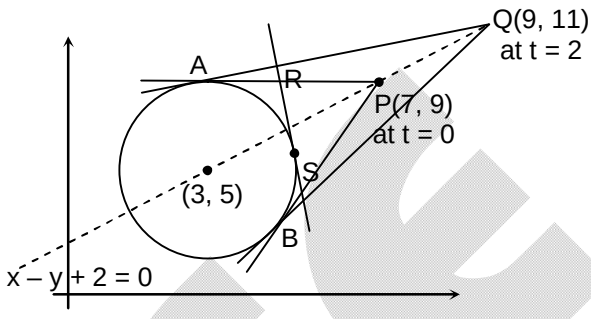
Line : $x - y + 2 = 0$

At t = 2 P is (9, 11)

At t = 0 perimeter of $\Delta PRS = 2AP = 4\sqrt{7}$

At t = 2 perimeter = $4\sqrt{17}$

a = 17 and b = 7



36. A, C, D

Sol. Locus of z^3 is a semi-circle with ω and ω^2 as end points of diameter

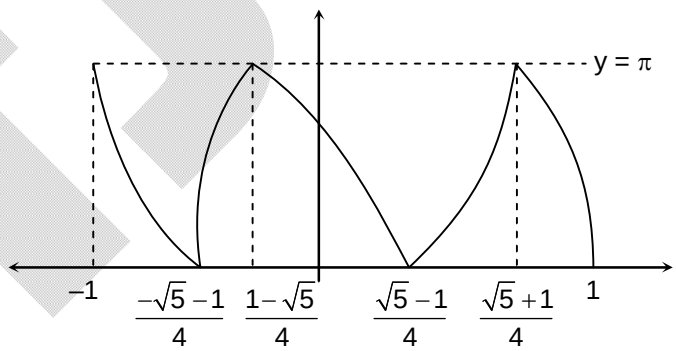
$|z + 2i||z + \sqrt{3} - i||z - \sqrt{3} - i| = |z^3 - 8i|$

So, its minimum value = $\frac{1}{2}\left(\sqrt{257} - \sqrt{3}\right)$

$\alpha = 257$ and $\beta = 3$

37. B, C, D

Sol. The graph of f(x) is



38. D

Sol. $(\tan^2\theta)^{1012} - {}^{2024}C_{2022}(\tan^2\theta)^{1011} + {}^{2024}C_{2020}(\tan^2\theta)^{1010} \ldots\ldots - {}^{2024}C_2\tan^2\theta + 1 = \frac{\cos(2024\theta)}{\cos^{2024}\theta}$

RHS = 0 at $\theta = (2m+1)\frac{\pi}{4048}$

So, $a = {}^{2024}C_{2022} = 1012 \cdot 2023$ and b = 1

39. C

Sol. $F \propto v^2 \Rightarrow ma = kv^2 \Rightarrow m\frac{dv}{dx} \cdot v = kv^2$

$\Rightarrow m\frac{dv}{v} = kdx \Rightarrow m\int_{500}^{100}\frac{dv}{v} = k\int_0^{0.2}dx \Rightarrow \frac{k}{m} = -5\log_e 5$

Now, $a = \frac{dv}{dt} = \frac{kv^2}{m} \Rightarrow \frac{dv}{v^2} = \frac{k}{m}dt$

$\Rightarrow \int_{500}^{100} \frac{dv}{v^2} = -5\log_e 5 \int_0^t dt \Rightarrow t = 0.0009941$ sec

So, $[10^4 \cdot t] = 9$

40. B

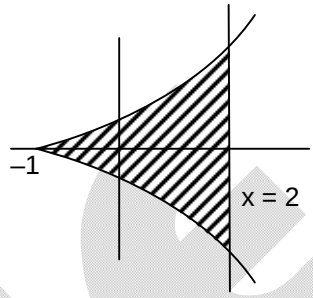
Sol. Let $\frac{dy}{dx} = p$

$y = px + p - p^3$; $y' = p + xp' + (1 - 3p^2)p'$

$\Rightarrow p^2 = \frac{x+1}{3}$

So, curve C is $27y^2 = 4(x + 1)^3$

Area = $2\int_{-1}^{2} \frac{2}{3\sqrt{3}}(x+1)^{\frac{3}{2}} dx = \frac{24}{5}$



41. C

Sol. $n(S) = 4^6$

Let E, W, N, S stand for 1 unit movement along +ve, –ve x-direction, y-directions

So, the sequence of steps is N N E E E W or N N N S E E

So, number of favourable cases = $\frac{6!}{2!3!} + \frac{6!}{2!3!} = 120$

$P(E) = \frac{120}{4^6} = \frac{15}{4^4 \cdot 2} = \frac{15}{512}$

42. B

Sol. L_1 and L_2 are coplanar

So, they are intersecting $\Rightarrow 6 - \lambda = 1 + \mu$ and $-1 + 2\lambda = -1 + 3\mu$

$\Rightarrow \lambda = 3$ and $\mu = 2$

$\cos\theta = \frac{(2\vec{b} - \vec{a}) \cdot (\vec{a} + 2\vec{b})}{|2\vec{b} - \vec{a}||\vec{a} + 3\vec{c}|} = \frac{1}{\sqrt{2}} \Rightarrow \theta = \frac{\pi}{4}$ so, $[\tan\theta] + 3 = 4$

Point of intersection is $3\vec{a} + 5\vec{b}$. It's distance from origin is $\sqrt{34} = d$

So, $[d] = 5$

43. C

Sol. (P) $2024^{2024} = 2^{6072} \cdot 11^{2024} \cdot 23^{2024}$

$f_1(2024^{2024}) = (2025)^2 \cdot 6073 = 3^8 \cdot 5^4 \cdot 6073$

$f_2(2024^{2024}) = f_1(3^8 \cdot 5^4 \cdot 6073) = 9 \cdot 5 \cdot 2 = 90$

$f_3(2024^{2024}) = f_1(90) = 12$

$f_4(2024^{2024}) = f_1(12) = 6$

$f_5(2024^{2024}) = f_1(6) = 4$

$f_6(2024^{2024}) = f_1(4) = 3$

(Q) $|A^{-1}| = -2 \Rightarrow |A| = -\frac{1}{2}$

$|adj(2A)| = |2A|^2 = (2^3|A|)^2 = 16$

$|2A| = -4$

$\Rightarrow \frac{1}{3}(|adj(2A)| + |2A|) = 4$

(R) A(a, b). The centre of the given circle (1, –2)
Equation of circumcircle of ΔABC is $(x - a)(x - 1) + (y - b)(y + 2) = 0$
$x^2 + y^2 - (a + 1)x + (2 - b)y - a - 2b = 0$ ..... (1)
$x^2 + y^2 - x + y - 2 = 0$ ..... (2)
Comparing equation (1) and (2), we get
$a + 1 = 1 \Rightarrow a = 0$, $2 - b = 1 \Rightarrow b = 1 \Rightarrow a + b = 1$

(S) $\sin\frac{\pi}{7}\sin\frac{2\pi}{7}\sin\frac{3\pi}{7} = \frac{\sqrt{7}}{8}$

$\cos\frac{\pi}{7}\cos\frac{2\pi}{7}\cos\frac{3\pi}{7} = \frac{1}{8}$

$\Rightarrow \tan\frac{\pi}{7}\tan\frac{2\pi}{7}\tan\frac{3\pi}{7} = \sqrt{7}$

44. D

Sol. (P) $\because$ α, β, γ are roots of $x^3 - x^2 - 2x - 3 = 0$

$\sum\alpha = 1$, $\sum\alpha\beta = -2$, $\alpha\beta\gamma = 3$

$$\Delta = \begin{vmatrix} 2\beta\gamma - \alpha^2 & \gamma^2 & \beta^2 \\ \gamma^2 & 2\alpha\gamma - \beta^2 & \alpha^2 \\ \beta^2 & \alpha^2 & 2\alpha\beta - \gamma^2 \end{vmatrix} = \begin{vmatrix} \alpha & \beta & \gamma \\ \beta & \gamma & \alpha \\ \gamma & \alpha & \beta \end{vmatrix}^2$$

$= \left(\sum\alpha\right)^2 \cdot \left(\sum\alpha^2 - \sum\alpha\beta\right)^2 = 1 \cdot \left(\left(\sum\alpha\right)^2 - 3\sum\alpha\beta\right)^2 = 49$

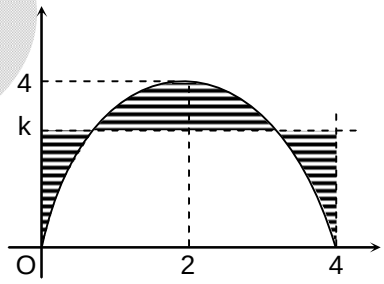
$\sqrt{\Delta} = 7$

(Q) $f(x) = x(4 - x)$ and $y = k$
F(k) is the shaded regions area

Area $A = 2\left[\int_0^k f^{-1}(x)\,dx + \int_k^4 \left(2 - f^{-1}(x)\right)dx\right]$

$\frac{dA}{dk} = 2\left(f^{-1}(k) - 2 + f^{-1}(k)\right)$

$\frac{dA}{dk} = 0 \Rightarrow f^{-1}(k) = 1 \Rightarrow k = 3$

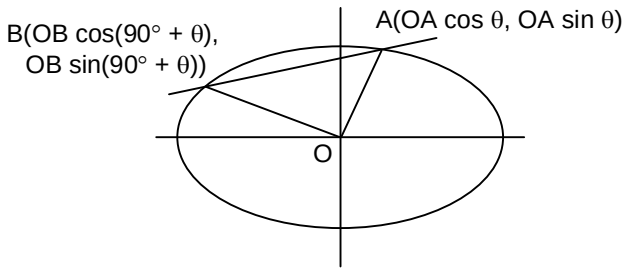


(R) $\frac{1}{OA^2} = \frac{\cos^2\theta}{9} + \frac{\sin^2\theta}{\lambda^2}$ ..... (1)

$\frac{1}{OB^2} = \frac{\sin^2\theta}{9} + \frac{\cos^2\theta}{\lambda^2}$ ..... (2)

$\because$ $\frac{1}{OA^2} + \frac{1}{OB^2} = \frac{14}{45}$

$\Rightarrow \frac{1}{\lambda^2} = \frac{14}{45} - \frac{1}{9} = \frac{1}{5} \Rightarrow \lambda^2 = 5$



(S) Hints: Use Sandwich Theorem

45. A

Sol. $g(x) = f^{-1}(x) \Rightarrow f(g(x)) = x$

$f'(g(x)) \cdot g'(x) = 1 \Rightarrow g'(x) = \frac{1}{f'(g(x))}$

and $g''(x) = -\frac{1}{\left(f'(g(x))\right)^2} \cdot f''(g(x)) \cdot g'(x) = -\frac{f''(g(x))}{\left(f'(g(x))\right)^3}$

(P) $f(x) = 2x + \cos x \quad f(0) = 1 \quad g(1) = 0$
$f'(x) = 2 - \sin x \quad f'(0) = 2$
$f''(x) = -\cos x \quad f''(0) = -1$

$$g''(1) = -\frac{f''(0)}{(f'(0))^3} = \frac{1}{8}$$

$$\Rightarrow \frac{4}{g''(1)} = 2$$

(Q) $f(x) = x^x$; $f(1) = 1 \Rightarrow g(1) = 1$
$f'(x) = x^x(1 + \log_e x) \Rightarrow f'(1) = 1$
$f''(x) = x^x(1 + \log_e x)^2 + x^{x-1} \Rightarrow f''(1) = 2$

$$g''(1) = -\frac{f''(1)}{(f'(1))^3} = -2$$

(R) $f(x) = \tan^{-1} x$; $f(0) = 0 \Rightarrow g(0) = 0$

$$f'(x) = -\frac{1}{1+x^2} \text{ ; } f'(0) = 1$$

$$f''(x) = -\frac{2x}{(1+x^2)^2} \Rightarrow f''(0) = 0$$

$$g''(0) = -\frac{f''(0)}{(f'(0))^3} = 0$$

(S) $f(x) = e^{x^3+x}$ $f(0) = 1 \Rightarrow g(1) = 0$

$$f'(x) = e^{x^3+x}(3x^2+1) \Rightarrow f'(0) = 1$$

$$f''(x) = e^{x^3+x}(3x^2+1) + 6xe^{x^3+x} \Rightarrow f''(0) = 1$$

$$\Rightarrow g''(1) = -\frac{f''(0)}{(f'(0))^3} = -1$$

SECTION – B

46. 2

Sol. The series becomes $S = \frac{1-2\cdot\left(\frac{2}{3}\right)}{1-\frac{2}{3}+\left(\frac{2}{3}\right)^2} + \frac{2\cdot\left(\frac{2}{3}\right)-4\cdot\left(\frac{2}{3}\right)^3}{1-\left(\frac{2}{3}\right)^2+\left(\frac{2}{3}\right)^4} + \frac{4\left(\frac{2}{3}\right)^3-8\left(\frac{2}{3}\right)^7}{1-\left(\frac{2}{3}\right)^4+\left(\frac{2}{3}\right)^8} + ...$

Put $\frac{2}{3} = x$

$$S = \frac{1-2x}{1-x+x^2} + \frac{2x-4x^3}{1-x^2+x^4} + \frac{4x^3-8x^7}{1-x^4+x^8} + ... \infty = \frac{2x+1}{1+x+x^2}$$

At $x = \frac{2}{3}$; $S = \frac{21}{19} = \frac{3\cdot 7}{19}$

47. 5

Sol. Using Leibnitz Theorem, we get $\frac{d^{10}y}{dx^{10}} = (90 - 3x - x^2)\sin x + (20x + 30)\cos x$

$P'(x) = -3 - 2x$
$|P'(1)| = |-5| = 5$

48. 4

Sol. $I_1 = \int_0^{\infty} \left(\frac{\log_e t}{1-t}\right)^2 dt$ [Substituting $\tan\theta = t$]

$I_2 = \int_0^1 \frac{\log_e (1-t)}{t} dt$ [Substituting $\sin\theta = t$]

$I_1 = \int_0^1 \left(\frac{\log_e t}{1-t}\right)^2 dt + \int_1^{\infty} \left(\frac{\log_e t}{1-t}\right)^2 dt$

Using $t = \frac{1}{z}$ in 2nd integral, we get

$= -4\int_0^1 \frac{\log_e t}{1-t} dt = 4I_2 \Rightarrow \frac{I_1}{I_2} = 4$

49. 6

Sol. Solving determinant, we get $8 \cdot \sum \cos A = 6 \cos A \cos B \cos C$

So, $8 \cdot \sum \sec A \sec B = 6$

50. 1

Sol. f(x) is periodic with period = 1

$n = 2024 \times 3 \times \frac{1}{2} = 3036 = 2^2 \cdot 3 \cdot 11 \cdot 23$

$m = 3036\left(1-\frac{1}{2}\right)\left(1-\frac{1}{3}\right)\left(1-\frac{1}{11}\right)\left(1-\frac{1}{23}\right) = 2640$

So, $\left[\frac{m}{2024}\right] = 1$

51. 3

Sol. The equation can be factorized as $(x - 1)(x + 2)(x^2 + (2a - 1)x + 1) = 0$

$(2a - 1)^2 - 4 < 0 \Rightarrow -\frac{1}{2} < a < \frac{3}{2} \Rightarrow a = 1$

So, $P = \frac{1}{6}$